What's more to come?

If you haven't come across the moniker James Webb Space Telescope, you are living under a rock. Currently, the JWST is the pinnacle of technology advancement in the domain of astrophysics. The telescope has managed to expand the area of research for astronomers with discoveries such as the Eagle Nebula Pillars, galaxies as old as 13.4 billion years ago and to get the first direct image of an exoplanet. The telescope has opened a new avenue for astronomers to procure and preserve data about the universe.

Having said that, the James Webb Space Telescope extensively uses artificial intelligence to analyse data from deep space through the machine learning model Morpheus. The model is trained on Nvidia GPUs and is utilised to analyse the generated data from JWST.

"The power here is that we used exclusively synthetic telescope data generated by computer simulations to train this AI, and then applied it to real telescope data. This has never been done before in our field, and paves the way for a deluge of discoveries as James Webb Telescope data rolls in." claimed Cassandra Hall, assistant professor of astrophysics, principal investigator of the Exoplanet and Planet Formation Research Group.

Jason Terry stated, "The potential for good data is exploding, so it's a very exciting time for the field." and indeed it is. With JWST, scientists can explore exoplanetary systems irrespective of their extremely bright, high resolution, forming environments. The greatest example of this would be the image that JWST captured of Eagle Nebula where the telescope has used its infra technology to capture even the clouds that are forming around the young stars. (below)

Particularly, when we speak about exoplanets artificial intelligence is playing a pivotal role in new generation devices such as the Atacama Large Millimeter/submillimeter Array (ALMA) to analyse the formation of the planets in the protoplanetary accretion disks. These devices are the future of astrophysics research and are capable of their discoveries because of the alignment of AI in their systems. Or else, imagine having to find an exoplanet in a thick layer of gas, thicker than the one between Earth and Sun with this advanced technology.

Moreover, these are fragments of what AI is doing for space exploration. The future now holds great opportunities in the discovery of the universe and the upcoming being NASA's Parker Solar Probe mission which is scheduled for 2024. The mission aims at the probe reaching the outer atmosphere of the Sun. Sounds simple, not at all. The probe is expected to be off 4 million miles of the Sun's surface which requires it to have the tolerance for 2500°F (or 1370 °C) temperature. In addition, the probe will be equipped with a magnetometer and an imaging spectrometer. This will help us determine how the Sun interacts with the planets in the solar system and we can detect the solar storms on Earth better. AI is nowhere limited to generating data based on prompts. It is the solution for a lot more problems that affect our understanding of the universe.

Additional Reading

- https://dgit.in/AIExo
- https://dgit.in/KinEDV
- https://dgit.in/NewP
- https://dgit.in/BHole
- https://dgit.in/JWST12
- https://dgit.in/AIsp

"When we applied our models to a set of older observations, they identified a disk that wasn't known to have a planet despite having already been analyzed. Like previous discoveries, we ran simulations of the disk and found that a planet could re-create the observation."

The AI tool in question works on the speed and efficiency of discovering forming planets especially when the telescopes deliver a flood of data on exoplanets in the Milky Way. The tool also works on identifying a signal in the previously analysed data making sure that there are no planets gone undiscovered.

"This demonstrates that our models and machine learning, in general, have the ability to quickly and accurately identify important information that people can miss" stated Terry.

There are AI models that are developing to underline evolution and formation of planetary systems. In a nutshell, astronomers can now discover planets that have been previously undetectable with traditional methods.

AI in astronomy has the potential of discovering the obscure truths of the universe and this will only intensify with the advancements in technology. What are your thoughts about AI's contribution in space exploration and celestial bodies' discovery, Let us know at editor@digit.in